

EA-0013 - AQELYN Engineering Archive

AQELYN - EA-0013 Engineering Archive

IS-013 - AQELYN Risk Intelligence Engine

****Archive ID:**** EA-0013

****Implementation Specification:**** IS-013

****Component:**** AQELYN Risk Intelligence Engine

****Project:**** AQELYN

****System Type:**** Cyber Security Operating Environment

****Status:**** COMPLETE

****Repository Impact:**** No top-level repository structure changes

****Breaking Changes:**** None

****Engineering Phase:**** Phase 3

****Predecessor Archives:**** EA-0001 through EA-0012

****Next Specification:**** IS-014

Document Control

| Field | Value |

---|---

| Document | Engineering Archive EA-0013 |

| Specification | IS-013 - AQELYN Risk Intelligence Engine |

| Publication Format | Markdown, PDF, HTML, ZIP |

| Source of Truth | MD/EA-0013.md |

| Archive Rule | Implementation Specification -> Engineering Archive -> Continue |

| Repository Rule | Fixed repository structure; no redesign |

| Completion State | IS-013 complete; EA-0013 generated |

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

| Engineering Archive | Implementation Specification | Component |

---|---|---

| EA-0001 | IS-001 | AQELYN Kernel |

| EA-0002 | IS-002 | Universal Object Model |

| EA-0003 | IS-003 | AQELYN Event Bus |

EA-0004	IS-004	AQELYN Evidence Engine
EA-0005	IS-005	AQELYN Knowledge Graph
EA-0006	IS-006	AQELYN Trust Engine
EA-0007	IS-007	AQELYN Mission Engine
EA-0008	IS-008	AQELYN Workflow Engine
EA-0009	IS-009	AQELYN Policy Engine
EA-0010	IS-010	AQELYN Compliance & Governance Engine
EA-0011	IS-011	AQELYN Identity & Access Governance Engine
EA-0012	IS-012	AQELYN Asset & Configuration Governance Engine
EA-0013	IS-013	AQELYN Risk Intelligence Engine

The fixed repository structure remains:

```
AQELYN/  
■■■ archive/  
■■■ blueprint/  
■■■ docs/  
■■■ src/  
■■■ tests/  
■■■ tools/  
■■■ build/  
■■■ releases/  
■■■ scripts/  
■■■ assets/  
■■■ examples/  
■■■ plugins/  
■■■ sdk/  
■■■ api/  
■■■ README.md
```

The engineering rule remains:

```
Finish Implementation Specification  
↓  
Generate Engineering Archive  
↓  
Continue
```

No Engineering Archive may be skipped.

2. IS-013 Specification Identity

```
Specification ID: IS-013  
Name: AQELYN Risk Intelligence Engine  
Engineering Archive Target: EA-0013  
Project: AQELYN  
System Type: Cyber Security Operating Environment  
Status: Complete  
Predecessor: IS-012 - AQELYN Asset & Configuration Governance Engine
```

3. Purpose

The AQELYN Risk Intelligence Engine transforms technical findings, governance results, configuration state, mission impact, and threat intelligence into a unified, evidence-backed view of organizational cyber risk.

Unlike a traditional risk register, the AQELYN Risk Intelligence Engine continuously evaluates risk based on live operational data from across the Cyber Security Operating Environment.

It answers:

What are the highest organizational cyber risks?
Which risks threaten current missions?
Which risks originate from configuration drift?
Which risks originate from identity weaknesses?
Which assets contribute most to enterprise risk?
Which policies reduce or increase risk?
How has risk changed over time?
Which evidence supports every risk assessment?
Which remediation actions reduce risk most effectively?

4. Mission

The engine shall provide:

Enterprise risk modeling
Operational risk assessment
Mission risk evaluation
Asset risk scoring
Identity risk scoring
Policy effectiveness analysis
Configuration risk analysis
Threat intelligence correlation
Evidence-backed risk calculations
Continuous risk monitoring
Risk prioritization
Risk trend analysis
Risk forecasting
Executive risk reporting

5. Scope

5.1 In Scope

Risk catalog
Risk register
Risk scoring
Risk aggregation
Mission risk
Asset risk
Identity risk
Configuration risk
Compliance risk
Threat correlation
Business impact analysis
Evidence-backed assessments
Risk ownership
Risk acceptance
Risk treatment planning
Risk monitoring
Risk reporting

5.2 Out of Scope

Financial accounting systems
Enterprise budgeting
Insurance management
Business continuity execution
Disaster recovery orchestration
Physical safety management
Legal case management

6. Dependencies

IS-013 depends on the previously completed AQELYN engines:

```

IS-001 AQELYN Kernel
IS-002 Universal Object Model
IS-003 AQELYN Event Bus
IS-004 AQELYN Evidence Engine
IS-005 AQELYN Knowledge Graph
IS-006 AQELYN Trust Engine
IS-007 AQELYN Mission Engine
IS-008 AQELYN Workflow Engine
IS-009 AQELYN Policy Engine
IS-010 AQELYN Compliance & Governance Engine
IS-011 AQELYN Identity & Access Governance Engine
IS-012 AQELYN Asset & Configuration Governance Engine

```

7. High-Level Architecture

```

AQELYN Risk Intelligence Engine
■
■■■ Risk Registry
■■■ Risk Assessment Engine
■■■ Risk Correlation Engine
■■■ Threat Intelligence Correlator
■■■ Mission Impact Analyzer
■■■ Risk Scoring Engine
■■■ Risk Trend Analyzer
■■■ Risk Forecast Engine
■■■ Risk Treatment Manager
■■■ Risk Evidence Binder
■■■ Executive Reporting Service
■■■ Risk Event Publisher

```

8. Functional Requirements

FR-013-001 - Risk Registry

The engine shall maintain an authoritative registry of all identified cyber risks.

Each risk shall include:

```

risk_id
title
description
category
severity
likelihood
impact
owner
status
evidence_links
created_at
updated_at

```

FR-013-002 - Risk Scoring

The engine shall calculate dynamic risk scores using inputs from:

```

Asset criticality
Configuration drift
Identity governance
Mission dependency
Policy compliance
Threat intelligence
Trust score
Evidence confidence
Historical observations

```

FR-013-003 - Risk Categories

Supported categories include:

- Operational Risk
- Mission Risk
- Identity Risk
- Asset Risk
- Configuration Risk
- Compliance Risk
- Threat Risk
- Supply Chain Risk
- Third-Party Risk
- Strategic Risk

FR-013-004 - Risk Ownership

Each risk shall have:

- Business Owner
- Technical Owner
- Security Owner
- Mission Owner
- Review Frequency
- Treatment Status

FR-013-005 - Evidence Binding

Every calculated risk shall reference immutable evidence stored in the AQELYN Evidence Engine.

No risk assessment shall exist without traceable supporting evidence.

FR-013-006 - Risk Lifecycle

Each risk progresses through:

- Identified
- Validated
- Assessed
- Accepted
- Mitigated
- Monitored
- Closed
- Archived

FR-013-007 - Risk Treatment

The engine shall support:

- Accept
- Avoid
- Reduce
- Transfer
- Monitor
- Escalate

FR-013-008 - Continuous Monitoring

The engine shall continuously re-evaluate risks when events occur in:

- Asset Governance
- Identity Governance
- Compliance
- Mission Engine
- Policy Engine
- Threat Intelligence
- Configuration Drift
- Workflow Engine

FR-013-009 - Risk Trends

Historical risk data shall support:

- Trend analysis
- Forecasting
- Baseline comparison
- Seasonal analysis
- Mission readiness impact

FR-013-010 - Executive Reporting

The engine shall produce dashboards and reports including:

- Enterprise Risk Overview
- Top Risks
- Mission Risk Summary
- Risk by Business Unit
- Risk by Asset Type
- Risk Trends
- Treatment Progress
- Compliance Impact

9. Non-Functional Requirements

The engine shall provide:

- Continuous evaluation
- Evidence-backed decisions
- Auditability
- Explainability
- Horizontal scalability
- Event-driven synchronization
- Policy integration
- Repository stability
- Backward compatibility

10. Core Risk Intelligence Flow

```
graph TD
    A[Operational Event] --> B[Risk Detection]
    B --> C[Evidence Collection]
    C --> D[Threat Correlation]
    D --> E[Risk Scoring]
    E --> F[Mission Impact Analysis]
    F --> G[Treatment Recommendation]
    G --> H[Executive Reporting]
    H --> I[Risk Event Published]
```

11. Internal Component Architecture

The AQELYN Risk Intelligence Engine is composed of modular services that communicate through the AQELYN Event Bus.

```
AQELYN Risk Intelligence Engine
■
■■■ Risk Registry
■■■ Risk Assessment Engine
■■■ Risk Correlation Engine
■■■ Threat Intelligence Correlator
■■■ Mission Impact Analyzer
■■■ Risk Scoring Engine
■■■ Risk Trend Analyzer
■■■ Risk Forecast Engine
■■■ Risk Treatment Manager
■■■ Risk Evidence Service
■■■ Executive Reporting Service
■■■ Risk Event Publisher
```

Each component shall operate independently while sharing the Universal Object Model and Evidence Engine.

12. Component Specifications

12.1 Risk Registry

The Risk Registry is the authoritative repository for all governed cyber risks.

Responsibilities:

```
Risk creation
Risk identification
Risk ownership
Risk lifecycle
Risk relationships
Historical tracking
```

Supported risk sources:

```
Compliance findings
Asset governance
Identity governance
Threat intelligence
Configuration drift
Mission assessments
Policy violations
Workflow escalations
```

12.2 Risk Assessment Engine

Continuously evaluates risk using evidence-backed inputs.

Inputs include:

```
Likelihood
Business impact
Mission impact
Technical severity
Evidence confidence
Trust score
Exposure
Historical observations
```

Output:

```
Dynamic enterprise risk score
```

12.3 Risk Correlation Engine

Correlates related risks originating from different AQELYN engines.

Correlation dimensions:

- Shared assets
- Shared identities
- Shared missions
- Shared evidence
- Shared policies
- Shared vulnerabilities
- Shared business units

12.4 Threat Intelligence Correlator

Consumes external and internal threat intelligence.

Supported sources include:

- Threat feeds
- Indicators of Compromise
- Adversary profiles
- Attack techniques
- Campaign intelligence
- Security advisories

Threat intelligence shall enrich, not replace, internal evidence.

12.5 Mission Impact Analyzer

Calculates mission impact resulting from cyber risk.

Factors include:

- Mission dependency
- Mission priority
- Mission availability
- Asset dependency
- Identity dependency
- Critical business functions

12.6 Risk Scoring Engine

Calculates normalized risk scores.

Inputs:

- Likelihood
- Impact
- Exposure
- Criticality
- Compliance status
- Threat level
- Trust score
- Evidence quality

Output:

- Risk Score
- Risk Level
- Confidence Level

12.7 Risk Trend Analyzer

Maintains historical trends.

Capabilities:

- Trend analysis
- Historical comparison
- Risk growth detection
- Risk reduction tracking
- Seasonal analysis

12.8 Risk Forecast Engine

Supports predictive analysis.

Forecasts include:

- Expected risk growth
- Mission degradation
- Compliance deterioration
- Exposure increase
- Treatment effectiveness

12.9 Risk Treatment Manager

Supports governance of remediation activities.

Treatment states:

- Planned
- Approved
- In Progress
- Completed
- Rejected
- Deferred

12.10 Risk Evidence Service

Maintains immutable evidence references for every calculated risk.

Evidence sources include:

- Compliance evaluations
- Configuration snapshots
- Asset inventory
- Identity governance
- Threat intelligence
- Workflow approvals
- Policy evaluations
- Mission assessments

13. Universal Object Model Extensions

The following objects extend IS-002.

13.1 Risk

- Risk:
- risk_id
- category
- likelihood
- impact
- score
- confidence
- owner
- status

13.2 RiskAssessment

```
RiskAssessment:  
assessment_id  
risk_id  
assessment_date  
methodology  
calculated_score  
evidence
```

13.3 RiskTreatment

```
RiskTreatment:  
treatment_id  
risk_id  
strategy  
owner  
due_date  
status
```

13.4 RiskTrend

```
RiskTrend:  
trend_id  
risk_id  
observation_date  
score
```

13.5 RiskRelationship

```
RiskRelationship:  
relationship_id  
source_risk  
target_risk  
relationship_type
```

14. Knowledge Graph Integration

Relationships include:

```
Risk  
↓  
affects  
↓  
Mission  
  
Risk  
↓  
originates_from  
↓  
Asset  
  
Risk  
↓  
owned_by  
↓  
Identity  
  
Risk  
↓  
governed_by  
↓  
Policy  
  
Risk  
↓  
supported_by  
↓  
Evidence
```

15. Event Bus Integration

The engine publishes standardized events.

15.1 Risk Events

```
risk.created  
risk.updated  
risk.closed  
risk.archived
```

15.2 Assessment Events

```
risk.assessed  
risk.recalculated  
risk.score.changed
```

15.3 Treatment Events

```
risk.treatment.started  
risk.treatment.completed  
risk.accepted  
risk.escalated
```

15.4 Forecast Events

```
risk.forecast.updated  
risk.trend.changed  
mission.risk.changed
```

16. Evidence Engine Integration

The Risk Intelligence Engine references immutable evidence from:

```
Compliance findings  
Asset evidence  
Configuration evidence  
Identity evidence  
Mission evidence  
Workflow evidence  
Threat evidence
```

Evidence identifiers shall never be modified after creation.

17. Policy Engine Integration

The Policy Engine supplies:

```
Risk acceptance policies  
Risk scoring rules  
Treatment requirements  
Escalation policies  
Review frequency
```

18. Compliance Integration

IS-010 provides:

```
Compliance findings  
Control effectiveness  
Assessment results
```

Regulatory violations
Exception records

These become inputs to enterprise risk calculations.

19. Identity Governance Integration

IS-011 contributes:

Privileged identities
Identity anomalies
Access governance findings
Ownership information

Identity weaknesses increase calculated risk.

20. Asset Governance Integration

IS-012 provides:

Asset inventory
Configuration drift
Criticality
Exposure
Ownership
Baseline compliance

Asset state is a primary contributor to operational risk.

21. Mission Engine Integration

Mission Engine contributes:

Mission dependency
Mission priority
Mission availability
Mission objectives

Mission impact directly influences enterprise risk prioritization.

22. Workflow Engine Integration

Workflow supports:

Risk review
Risk approval
Treatment workflow
Exception approval
Escalation workflow

23. Trust Engine Integration

Trust Engine contributes:

```
Evidence confidence
Asset trust
Identity trust
Configuration trust
```

These values influence confidence in calculated risk scores.

24. Public APIs

24.1 Risk API

```
GET /risks
POST /risks
GET /risks/{id}
PUT /risks/{id}
```

24.2 Assessment API

```
GET /risk-assessments
POST /risk-assessments
GET /risk-assessments/{id}
```

24.3 Treatment API

```
GET /risk-treatments
POST /risk-treatments
PUT /risk-treatments/{id}
```

24.4 Reporting API

```
GET /risk-dashboard
GET /risk-trends
GET /mission-risk
```

24.5 Forecast API

```
GET /risk-forecast
POST /risk-forecast/recalculate
```

25. Repository Impact

Implementation shall reside within the existing repository without modifying the approved structure.

```
AQELYN/
■■■ src/
■ ■■■ risk_intelligence/
■■■ tests/
■ ■■■ risk_intelligence/
■■■ docs/
■ ■■■ risk_intelligence/
■■■ api/
■ ■■■ risk_intelligence/
■■■ archive/
```

No top-level repository changes are permitted.

26. Security Architecture

The AQELYN Risk Intelligence Engine is a Tier-1 analytical governance subsystem responsible for producing evidence-backed enterprise cyber risk intelligence.

Every calculated risk shall be:

- Evidence-backed
- Explainable
- Auditable
- Policy-governed
- Version controlled
- Traceable
- Continuously re-evaluated

No risk assessment shall exist without supporting evidence.

26.1 Security Principles

The engine shall implement:

- Zero Trust
- Evidence Integrity
- Least Privilege
- Defense in Depth
- Continuous Validation
- Immutable Risk History
- Explainable Analytics
- Separation of Duties
- Security by Design

26.2 Risk Authorization Model

Only authorized governance roles may create, approve, or modify risk records.

Supported governance roles:

Role	Responsibility
Risk Administrator	Risk lifecycle management
Security Analyst	Risk assessment
Compliance Officer	Regulatory validation
Mission Owner	Mission impact approval
Business Owner	Business risk ownership
Executive Reviewer	Enterprise approval
Automation Service	Automated reassessment

Authorization decisions shall be enforced through the AQELYN Policy Engine.

26.3 Risk Integrity

Every risk record shall maintain:

- Risk Identifier
- Version
- Evidence References
- Assessment History
- Scoring Methodology
- Ownership
- Treatment History
- Review History

Historical assessments shall remain immutable.

26.4 Risk Evidence Protection

Risk evidence shall support:

- Immutable storage
- Evidence versioning
- Integrity verification
- Trust scoring
- Evidence lineage
- Audit history

Risk calculations shall never reference mutable evidence.

27. Risk Lifecycle

Every governed cyber risk follows a controlled lifecycle.

```
graph TD
    Identified --> Validated
    Validated --> Assessed
    Assessed --> Prioritized
    Prioritized --> Assigned
    Assigned --> Mitigated
    Mitigated --> Reviewed
    Reviewed --> Closed
    Closed --> Archived
```

27.1 Risk Assessment Lifecycle

```
graph TD
    EvidenceCollected[Evidence Collected] --> Correlation
    Correlation --> RiskScoring[Risk Scoring]
    RiskScoring --> MissionAnalysis[Mission Analysis]
    MissionAnalysis --> PolicyEvaluation[Policy Evaluation]
    PolicyEvaluation --> Approval
    Approval --> Monitoring
```

27.2 Risk Treatment Lifecycle

```
graph TD
    Planned --> Approved
    Approved --> Implemented
    Implemented --> Validated
    Validated --> Verified
    Verified --> Closed
```

28. Risk Evaluation Model

Risk calculations shall evaluate:

- Likelihood
- Business Impact
- Mission Impact
- Asset Criticality
- Identity Exposure
- Configuration Drift
- Threat Intelligence
- Policy Compliance
- Trust Score
- Evidence Confidence
- Historical Trends

28.1 Risk Levels

- Informational
- Low
- Moderate
- High
- Critical

Critical risks shall trigger immediate governance workflows.

29. Continuous Risk Intelligence

The engine shall continuously monitor:

- Configuration drift
- Asset exposure
- Identity governance
- Compliance findings
- Threat intelligence
- Mission changes
- Trust score changes
- Policy violations
- Workflow escalations
- Evidence updates

Every significant change shall trigger automatic reassessment.

30. Executive Reporting

30.1 Executive Reports

- Enterprise Risk Dashboard
- Top Enterprise Risks
- Mission Risk Summary
- Strategic Risk Trends
- Risk Reduction Progress

30.2 Operational Reports

- Open Risks
- Critical Risks
- Risk Treatments
- Asset Risk Distribution
- Identity Risk Distribution

30.3 Compliance Reports

- Compliance Risk
- Control Effectiveness
- Regulatory Risk
- Exception Tracking

30.4 Engineering Reports

Risk Correlation Graph
Risk Trend History
Evidence Coverage
Scoring Statistics
Forecast Accuracy

31. Failure Handling

31.1 Missing Evidence

Status:
Assessment Blocked

Action:
Collect additional evidence

31.2 Missing Owner

Status:
Governance Incomplete

Action:
Escalate ownership assignment

31.3 Calculation Failure

Status:
Assessment Failed

Action:
Retry according to risk calculation policy

31.4 Event Bus Failure

Risk events shall be queued until successful delivery.

31.5 External Threat Feed Failure

The engine shall continue operating using existing evidence and mark threat intelligence confidence accordingly.

32. Performance Requirements

The engine shall support:

Continuous enterprise reassessment
Large-scale risk calculations
Incremental recalculation
Parallel scoring
Real-time event ingestion
Executive dashboard generation

Risk calculations shall not interrupt operational AQELYN services.

33. Scalability Requirements

The architecture shall support:

- Millions of risk observations
- Millions of evidence references
- Large enterprise deployments
- Multi-cloud environments
- Hybrid infrastructures
- Distributed processing
- Multi-tenant operation

34. Testing Strategy

34.1 Unit Testing

Validate:

- Risk Registry
- Assessment Engine
- Correlation Engine
- Scoring Engine
- Trend Analyzer
- Forecast Engine
- Treatment Manager

34.2 Integration Testing

Verify interaction with:

- AQELYN Kernel
- Universal Object Model
- Event Bus
- Evidence Engine
- Knowledge Graph
- Trust Engine
- Mission Engine
- Workflow Engine
- Policy Engine
- Compliance Engine
- Identity Governance Engine
- Asset Governance Engine

34.3 System Testing

Validate end-to-end scenarios including:

- Risk creation
- Evidence collection
- Correlation
- Scoring
- Mission impact analysis
- Treatment workflow
- Executive reporting

34.4 Security Testing

Verify:

- Authorization
- Evidence integrity
- Risk integrity
- Audit trail completeness
- Policy enforcement
- Workflow security

34.5 Regression Testing

Ensure IS-001 through IS-012 continue operating without behavioral changes introduced by IS-013.

35. Acceptance Criteria

IS-013 shall be considered complete when:

```
Risk Registry is defined.  
Risk Assessment Engine is documented.  
Risk Correlation Engine is defined.  
Threat Intelligence integration is documented.  
Risk scoring model is defined.  
Risk lifecycle is documented.  
Treatment workflow is defined.  
Continuous monitoring is documented.  
Integration with IS-001 through IS-012 is complete.  
Repository structure remains unchanged.  
Testing strategy is documented.
```

36. Repository Validation

Implementation shall follow the approved repository structure.

```
AQELYN/  
■■■ src/risk_intelligence/  
■■■ tests/risk_intelligence/  
■■■ docs/risk_intelligence/  
■■■ api/risk_intelligence/  
■■■ archive/
```

No repository redesign is introduced.

37. Engineering Summary

The AQELYN Risk Intelligence Engine provides the analytical layer that transforms technical events into enterprise cyber risk intelligence.

It introduces:

```
Enterprise Risk Registry  
Dynamic Risk Assessment  
Threat Correlation  
Mission Impact Analysis  
Evidence-backed Risk Scoring  
Risk Forecasting  
Continuous Monitoring  
Executive Reporting  
Governance-driven Treatment Management
```

The engine integrates with all previous AQELYN components while preserving modularity and backward compatibility.

38. Specification Status

```
Specification ID : IS-013  
Title : AQELYN Risk Intelligence Engine  
Status : COMPLETE  
Engineering Archive : READY FOR GENERATION  
Next Artifact : EA-0013
```

Engineering workflow status:

```
EA-0001 COMPLETE
EA-0002 COMPLETE
EA-0003 COMPLETE
EA-0004 COMPLETE
EA-0005 COMPLETE
EA-0006 COMPLETE
EA-0007 COMPLETE
EA-0008 COMPLETE
EA-0009 COMPLETE
EA-0010 COMPLETE
EA-0011 COMPLETE
EA-0012 COMPLETE
IS-013 COMPLETE
EA-0013 READY FOR GENERATION
```

39. EA-0013 Engineering Objective

The objective of IS-013 was to introduce a dedicated Risk Intelligence Engine that enables AQELYN to continuously evaluate organizational cyber risk from evidence, threat intelligence, governance findings, mission impact, asset state, identity state, policy decisions, and compliance posture.

The engine extends AQELYN governance from individual controls, identities, and assets into enterprise risk intelligence and risk prioritization.

40. EA-0013 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

```
Risk registry management
Risk assessment
Risk correlation
Threat intelligence correlation
Mission impact analysis
Risk scoring
Risk trend analysis
Risk forecasting
Risk treatment management
Evidence binding for risk decisions
Executive reporting
Risk event publishing
```

The engine integrates with all previously completed AQELYN engines while preserving architectural modularity.

41. Major Engineering Decisions

41.1 Decision 1 - Dedicated Risk Intelligence Engine

Risk intelligence responsibilities are implemented as a standalone engine rather than embedded in Compliance, Policy, Asset Governance, or Mission.

Rationale:

```
Clear separation of risk analytics from governance source systems.
Independent lifecycle and scaling.
Better support for risk correlation and forecasting.
Improved traceability of enterprise risk decisions.
```

41.2 Decision 2 - Evidence-Backed Risk Model

No risk assessment shall exist without evidence links.

Benefits:

Risk decisions become auditable.
Risk scores are explainable.
Executive reporting can be traced to source evidence.
Compliance and mission risk can be supported by proof.

41.3 Decision 3 - Dynamic Risk Scoring

Risk scores are continuously recalculated from live operational signals rather than treated as static register values.

Benefits:

Risk posture reflects current operations.
Drift, exposure, identity, threat, and compliance changes affect risk.
Treatment effectiveness can be measured.
Mission readiness can consume live risk values.

41.4 Decision 4 - Event-Driven Risk Intelligence

Risk, assessment, treatment, forecast, trend, and mission risk state changes are published through the AQELYN Event Bus.

Examples include:

```
risk.created  
risk.updated  
risk.assessed  
risk.recalculated  
risk.score.changed  
risk.treatment.started  
risk.treatment.completed  
risk.forecast.updated  
risk.trend.changed  
mission.risk.changed
```

This maintains loose coupling between AQELYN engines.

41.5 Decision 5 - Universal Object Model Extension

New domain objects introduced include:

```
Risk  
RiskAssessment  
RiskTreatment  
RiskTrend  
RiskRelationship
```

These extend the Universal Object Model without modifying existing object definitions.

42. Architectural Integration Summary

| Engine | Integration |

|---|---|

| IS-001 Kernel | Runtime lifecycle and service registration |

| IS-002 Universal Object Model | Risk, assessment, treatment, trend, relationship objects |

| IS-003 Event Bus | Risk, assessment, treatment, trend, forecast, mission-risk events |

| IS-004 Evidence Engine | Evidence-backed risk calculations and assessment history |

| IS-005 Knowledge Graph | Risk, asset, identity, policy, mission, evidence relationships |

| IS-006 Trust Engine | Evidence confidence, asset trust, identity trust, configuration trust |

| IS-007 Mission Engine | Mission impact and mission risk prioritization |

| IS-008 Workflow Engine | Risk review, treatment, escalation, and approval workflows |

| IS-009 Policy Engine | Risk scoring rules, acceptance policies, escalation policies |

| IS-010 Compliance Engine | Compliance findings, control effectiveness, exceptions |

| IS-011 Identity Governance Engine | Privileged identities, access findings, ownership information |

| IS-012 Asset Governance Engine | Asset criticality, drift, exposure, ownership, baseline compliance |

No existing engine required redesign.

43. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/
■■■ src/risk_intelligence/
■■■ tests/risk_intelligence/
■■■ api/risk_intelligence/
■■■ docs/risk_intelligence/
■■■ archive/
```

No top-level directories were added, removed, or renamed.

44. Security Impact Summary

The specification introduces risk-intelligence-specific security controls:

```
Policy-driven risk authorization
Evidence-backed risk records
Immutable risk assessment history
Risk evidence protection
Separation of duties for risk acceptance and treatment approval
Traceable risk ownership
Continuous risk reassessment
Explainable risk analytics
```

No reduction in the security posture of existing components was identified.

45. Capabilities Added

The engine enables AQELYN to support:

```
Enterprise risk registry
Operational risk assessment
Mission risk evaluation
Asset risk scoring
Identity risk scoring
Policy effectiveness analysis
Configuration risk analysis
Threat intelligence correlation
Risk trend analysis
Risk forecasting
```

Risk treatment management
Executive risk reporting
Evidence-backed risk calculation
Continuous risk monitoring

46. Risks Identified

| Risk | Mitigation |

|---|---|

| Inaccurate risk scoring | Evidence-backed methodology and policy-defined scoring rules |

| Missing evidence | Assessment blocked state and evidence collection workflow |

| Threat feed quality issues | Confidence scoring and internal evidence priority |

| Over-correlation of risks | Knowledge Graph relationships and explainable correlation rules |

| Executive reporting misinterpretation | Traceable dashboards and evidence-linked summaries |

| Large-scale recalculation overhead | Incremental recalculation and parallel scoring |

| Unauthorized risk acceptance | Policy enforcement and separation of duties |

| Forecast uncertainty | Confidence levels and trend transparency |

No critical architectural risks were identified that require redesign.

47. Verification Summary

The specification defines verification for:

Unit testing
Integration testing
System testing
Security testing
Regression testing

Acceptance criteria cover risk registry, assessment engine, correlation engine, threat intelligence integration, scoring model, risk lifecycle, treatment workflow, continuous monitoring, integration with IS-001 through IS-012, repository validation, and testing documentation.

48. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

Modular architecture
Event-driven communication
Immutable evidence references
Traceability
Explainability
Security by design
Repository stability
Backward compatibility
Governance-before-archive discipline

49. Dependencies

Required:

EA-0001 through EA-0012
IS-001 through IS-012

Enables:

IS-014 and subsequent risk-dependent components

50. Completion Record

Engineering Archive : EA-0013
Implementation Specification : IS-013
Title : AQELYN Risk Intelligence Engine
Engineering Status : COMPLETE
Repository Status : UNCHANGED
Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED
Engineering Rule :
IS Completed
↓
EA Generated
↓
Continue

51. Archive Index Update

EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
EA-0008 IS-008 AQELYN Workflow Engine
EA-0009 IS-009 AQELYN Policy Engine
EA-0010 IS-010 AQELYN Compliance & Governance Engine
EA-0011 IS-011 AQELYN Identity & Access Governance Engine
EA-0012 IS-012 AQELYN Asset & Configuration Governance Engine
EA-0013 IS-013 AQELYN Risk Intelligence Engine

52. Engineering Phase Status

Completed Engineering Archives : EA-0001 through EA-0013

Current Status:
EA-0013 COMPLETE

Next Implementation Specification:
IS-014

EA-0013 is completed and archived. The engineering workflow is consistent with the project rule:

Implementation Specification -> Engineering Archive -> Continue

From this point onward, the next engineering artifact is IS-014.

53. Engineering Archive Publication Standard

EA-0013 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:


```
EA-xxxx/
■
■ ■ ■ ■ diagrams/
■ ■ ■ ■ Architecture.svg
■ ■ ■ ■ Component.svg
■ ■ ■ ■ Workflow.svg
■ ■ ■ ■ EventFlow.svg
■ ■ ■ ■ Integration.svg
■
■ ■ ■ ■ examples/
■ ■ ■ ■ example_*.md
■
■ ■ ■ ■ HTML/
■ ■ ■ ■ index.html
■ ■ ■ ■ styles.css
■ ■ ■ ■ assets/
■
■ ■ ■ ■ images/
■
■ ■ ■ ■ journal/
■ ■ ■ ■ Engineering_Journal.md
■
■ ■ ■ ■ manifest/
■ ■ ■ ■ manifest.json
■
■ ■ ■ ■ MD/
■ ■ ■ ■ EA-xxxx.md
■
■ ■ ■ ■ PDF/
■ ■ ■ ■ EA-xxxx.pdf
■
■ ■ ■ ■ requirements/
■ ■ ■ ■ Requirements_Matrix.md
■
■ ■ ■ ■ traceability/
■ ■ ■ ■ Traceability_Matrix.md
■
■ ■ ■ ■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

54. Requirements Matrix

Requirement ID	Requirement	Evidence in Archive	Status
FR-013-001	Maintain risk registry	Sections 8, 12	Complete
FR-013-002	Calculate dynamic risk scores	Sections 8, 12, 28	Complete
FR-013-003	Support risk categories	Section 8	Complete
FR-013-004	Maintain risk ownership	Sections 8, 26	Complete
FR-013-005	Bind risks to evidence	Sections 8, 16, 41	Complete
FR-013-006	Support risk lifecycle	Sections 8, 27	Complete
FR-013-007	Support risk treatment	Sections 8, 12, 27	Complete
FR-013-008	Continuously monitor risks	Sections 8, 29	Complete
FR-013-009	Support risk trends	Sections 8, 12	Complete
FR-013-010	Produce executive reporting	Sections 8, 30	Complete
NFR-013-001	Continuous evaluation	Sections 9, 29, 32	Complete
NFR-013-002	Evidence-backed decisions	Sections 9, 16, 41	Complete
NFR-013-003	Auditability	Sections 9, 26, 31	Complete
NFR-013-004	Explainability	Sections 9, 28, 41	Complete
NFR-013-005	Event-driven synchronization	Sections 9, 15, 41	Complete
NFR-013-006	Repository stability	Sections 25, 36, 43	Complete

55. Traceability Matrix

Source	Target	Relationship
IS-013 Purpose	EA-0013 Objective	Defines why the engine exists
Risk Registry	FR-013-001	Implements authoritative risk inventory
Risk Scoring Engine	FR-013-002	Implements dynamic risk scoring
Risk Categories	FR-013-003	Implements risk classification
Risk Ownership	FR-013-004	Implements accountable risk ownership
Risk Evidence Service	FR-013-005	Binds risk records to evidence
Risk Lifecycle	FR-013-006	Implements risk lifecycle
Risk Treatment Manager	FR-013-007	Implements treatment workflow
Continuous Risk Intelligence	FR-013-008	Implements continuous monitoring
Risk Trend Analyzer	FR-013-009	Supports historical trends
Executive Reporting Service	FR-013-010	Implements reporting
Evidence Engine Integration	Evidence-backed decisions	References immutable evidence
Policy Engine Integration	Scoring and acceptance rules	Determines risk policy behavior
Mission Engine Integration	Mission impact	Supports mission risk prioritization
Compliance Integration	IS-010	Supplies compliance findings
Identity Governance Integration	IS-011	Supplies identity risk context
Asset Governance Integration	IS-012	Supplies asset, drift, and exposure context
Event Bus Integration	NFR-013-005	Publishes risk events
Repository Validation	Repository Standard	Confirms no top-level redesign

56. Engineering Journal

Journal Entry - EA-0013

EA-0013 was created to archive completion of IS-013 - AQELYN Risk Intelligence Engine.

The archive records the expansion of AQELYN into dynamic cyber risk intelligence. IS-013 defines the structure needed to maintain governed risk records, calculate risk scores, correlate risks across AQELYN engines, bind risk decisions to evidence, analyze mission impact, manage treatments, forecast trends, and generate executive risk reporting.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

Risk intelligence must be modeled separately from compliance, asset governance, and mission readiness. The Risk Intelligence Engine consumes evidence and context from those engines but owns enterprise risk scoring, risk lifecycle, treatment governance, and risk reporting.

Governance Note

EA-0013 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

57. Examples

57.1 Example Risk

```
risk_id: RISK-0001
category: configuration_risk
likelihood: high
impact: critical
score: 92
confidence: 0.88
owner: security_owner
status: assessed
```

57.2 Example Risk Assessment

```
assessment_id: RA-1001
risk_id: RISK-0001
assessment_date: 2026-07-07T12:00:00Z
methodology: aqelyn_dynamic_risk_v1
calculated_score: 92
evidence:
- evidence://config-drift-asset-0002
- evidence://asset-criticality-asset-0002
- evidence://threat-intel-campaign-123
```

57.3 Example Risk Treatment

```
treatment_id: RT-2001
risk_id: RISK-0001
strategy: reduce
owner: technical_owner
due_date: 2026-08-15
status: in_progress
```

57.4 Example Risk Event

```
{
  "event_type": "risk.score.changed",
  "risk_id": "RISK-0001",
  "previous_score": 71,
  "new_score": 92,
  "reason": "Critical configuration drift on mission-critical asset",
  "source_engine": "aqelyn_risk_intelligence_engine"
}
```

58. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0013.md
PDF/EA-0013.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_risk_intelligence.md
```

59. Final Archive Statement

EA-0013 is the Engineering Archive for IS-013 - AQELYN Risk Intelligence Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

EA-0013 COMPLETE
IS-013 COMPLETE
NEXT: IS-014